

AMIRKHON GAIRATOV

Computer Vision Engineer

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SUMMARY

Computer Vision engineer with 1+ year of production experience building applied ML pipelines for surveillance, retail, and industrial clients — from dataset preparation and model fine-tuning through real-time inference and deployment. Background in biometric recognition, embedding-based systems, and time-dependent event pipelines, alongside independent research applying LLMs to real-time procedural systems.

EXPERIENCE

Computer Vision Engineer — Telecom Soft AI R&D

Tashkent, Uzbekistan

Applied CV / ML pipelines for surveillance, retail, and industrial clients

2025 – Present

- Delivered 10+ end-to-end computer vision projects, covering model fine-tuning, inference optimization, and production deployment on live camera feeds.
- Designed a best-shot face-crop selection algorithm for a biometric recognition pipeline, raising TAR@FAR=1e-4 from 0.28 to 0.95 without changing the underlying recognition model.
- Built stateful, time-dependent multi-camera event systems — tracking object and zone status over time and triggering downstream alerts based on configurable rules.
- Developed embedding-based recognition and re-identification systems, working with vector/embedding databases for face and person matching.
- Owned data science workstreams supporting model training: dataset review and curation, class balancing, and image quality enhancement.

PROJECTS

LLM as a Procedural Game Event Director — University Thesis Project

Unreal Engine 5, C++

- Built a soulslike game in UE5 where an LLM (GPT-4o-mini) selects real-time game events from player-state telemetry, returning structured JSON consumed by the game engine.
- Designed and ran an A/B study with 20 participants comparing the LLM-driven director against a deterministic rule-based baseline; defended findings in a full written thesis and presentation.

GoShiet — Student Class-Scheduling Platform

Team project

- Built a Django-based web platform for student class scheduling as part of a university team project.

File Manager

C, student project

- Implemented core file-system operations (navigation, file I/O, directory management) in C.

EDUCATION

Lomonosov Moscow State University — Tashkent Branch

Tashkent, Uzbekistan

B.S., Applied Mathematics and Informatics — Group M-23 (4th year)

Expected 2027

TECHNICAL SKILLS

- **Languages:** Python, C, C++, object-oriented design
- **Computer Vision / ML:** YOLO family, ByteTrack, BoT-SORT, RTMPose, PyTorch, ONNX, OpenCV, ArcFace, model fine-tuning & deployment
- **Other:** Unreal Engine 5 (C++), Django, embedding / vector databases